

WEEK 7 ANALYTICS TESTING & REFINEMENT REPORT

Prepared By: Abdullahi Ahmad

Track: Data Analytics

Project Phase: Week 7 (Testing & End-to-End Validation)

1. Dashboard Testing Approach & KPI Validation

To ensure the analytical outputs are reliable for HealthConnect Clinic decision-making, systematic testing was conducted on the Week 6 Power BI dashboard.

KPI Validation: The baseline dataset contains exactly 5,000 appointment records. The dashboard's primary "Total Appointments" KPI card successfully reflects this exact figure, confirming no data loss during the importation and transformation processes.

Filter & Slicer Testing: Cross-filtering functionality was tested by isolating specific reminder channels (e.g., WhatsApp) and adjusting demographic slicers (Gender and Age Group). All dependent charts and outcome-based logistical tables dynamically updated without producing calculation errors or visual breaks.

Validation Result: The dashboard accurately reflects the underlying dataset. The previously identified logistical findings—that clinic distance and wait times do not drive no-shows—remain mathematically consistent across all filtered demographic segments.

2. Mandatory HC-POD Cross-Track Testing & Refinement

To fulfill the end-to-end validation requirement, a collaborative testing cycle was executed with the Data Science track to validate the feature recommendations handed off in Week 6.

Track Collaborated With: Data Science.

Project Dependency: ML Model Feature Selection (validating the analytical insights against the baseline predictive model).

Component Tested: The baseline classification model predicting patient no-shows.

Information Provided: The Week 6 Data Science Handoff Memo recommending high weighting for booking_lead_days and reminder_channel.

Testing Activity Completed: A comparative A/B test of the model's predictive accuracy. The Data Science track ran the baseline model using standard demographics, followed by a re-test incorporating the analytically validated operational features (booking_lead_days, reminder_channel).

Issue / Finding Identified: The initial baseline model struggled with false negatives (predicting a patient would attend when they actually no-showed) because it lacked the operational context of advanced booking lead times.

Refinement / Action Taken: The Data Science track adjusted the algorithm to heavily weight booking_lead_days and reminder_channel as primary predictive nodes based on the Data Analytics EDA.

Retest Result: Incorporating these features significantly improved the model's recall for the "No-Show" class. The algorithm successfully identified high-risk appointments (e.g., those booked >25 days in advance with "None" for reminder channel) with much higher accuracy.

Validated Improvement (How it improved the overall solution): The predictive model is now mathematically grounded in actual patient behavior patterns rather than standard demographics. This ensures that any Week 8 AI or automated interventions will target the correct operational bottlenecks.